

AYUSH RANJAN

Software Engineer — Machine Learning Developer

📍 Ranchi, Jharkhand 📞 +91 9142115113 ✉ ayushranjan1271@gmail.com 🔗 LinkedIn

Summary

Aspiring Software Engineer and Machine Learning Developer with strong foundations in Python, Java, C++, and Data Structures & Algorithms. Experienced in ML classification systems, healthcare AI models, automated web scrapers, and cryptographic research. Author of a ternary logic-based S-Box cipher paper under academic review with interests in legal-tech and high-performance computing.

Skills

- Languages:** Python (Advanced), Java, C++, SQL, HTML/CSS
- Machine Learning:** Classification Models, Feature Engineering, Model Training, Model Evaluation, NLP
- Core CS:** Data Structures & Algorithms (300+ problems), OOPS, DBMS
- Data & Research:** Data Preprocessing, Feature Selection, Model Optimization, Cryptographic Analysis
- Tools:** AWS, Git/GitHub, Tkinter, Selenium, BeautifulSoup, Streamlit

Experience

Freelancing Club, VIT Bhopal

Core Member (Machine Learning)

Jan 2025 – Present

Bhopal, India

- Developed and delivered machine learning solutions as part of the core technical team.
- Collaborated with clients to automate data analysis and model-driven workflows.

Codetech

Data Science Intern

Nov 2024 – Jan 2025

Remote

- Implemented predictive models and optimized training pipelines across datasets.
- Improved model accuracy through feature tuning and analytical validation.

Cognifyz Technologies

Data Analysis Intern

Nov 2024 – Dec 2024

Remote

- Conducted statistical analysis and visualization for business insights.
- Streamlined data cleaning workflows for high-quality BI outputs.

1Stop.ai

Machine Learning Intern

Jan 2024 – Apr 2024

Remote

- Built ML models under strict project constraints and supervision.
- Assisted deployment of classification systems with robust edge-case handling.

Technical Projects

- Legal Document Classification (Python, NLP):** Classified 5,000+ legal documents using TF-IDF and SVM with 93% accuracy; reduced manual effort by 70%.
- Ternary Logic-Based S-Box Cryptography (Research):** Designed ternary logic encryption; validated security via NPCR, UACI, and entropy analysis; paper under review.
- Bin2Book – Web Scraper & GUI:** Developed multi-threaded Tkinter application to scrape and convert novels into PDFs using Selenium.

Achievements & Contributions

- Open-source contributor to SunPy and participant in Social Summer of Code (SSOC 2025).
- NSS Volunteer:** Co-organized blood donation camp (400+ units) and plantation drives (10,000+ trees).
- Certifications:** Deloitte Data Analytics Simulation, Python & Java Certificates, AWS Cloud.

Education

- Integrated M.Tech in Computer Science (Computational and Data Science), VIT Bhopal University (2023 – 2028)